

higher technologies. UXM of Keysight is such a device. However, options like this are expensive. Many developers who are creating a new RF system may look for something more economical. Moderate systems that measure frequencies below 6GHz are available for the purpose. Instruments like signal generators are used to generate signals and test the receiver of the IoT. Signal analysers take signals from the IoT transmitter and analyse these. All these come as handheld devices. These can be used by manufacturers as well as R&D departments."

Testing the battery capacity and power consumption of an IoT system is the third major criteria. Mombasawala continues, "We also have battery drain analysers. Battery drain testing is generally done at R&D stage and requires a voltage-current measurement setup."



Fig. 1: Keysight E6640A EXM Wireless



Fig. 2: Tektronix real-time spectrum analyser

Test device upgrades

T&M providers are trimming the tests supported by their equipment down to the transmitter, receiver and power-based measurements. They are making the test devices more modular. The port density is increas-

ing. Newer test devices support higher number of parallel devices under test (DUT) and improved test capacities at economical rates. This is making the T&M device hardware futureproof.

Typically, IoT systems are made up of at least one sensor, a microprocessor and an RF chip. The system draws and releases current of varying magnitudes in different states. So the optimal test equipment should come with dynamic current sensitivity ranges. These should also support sufficient bandwidth for proper signal sampling. Test equipment for IoT systems are being modified accordingly. Firm-

ware and software upgrades, hardware miniaturisation and capacity improvements are some other trends.

T&M players like Anritsu and Litepoint have rolled out new hardware for test devices intended for use in manufacturing floors. The modifica-

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